24. Simple Algebra

Year 4 · **Supplementary Worksheet** — extra practice & worked breakdowns

Worked Examples

Example 1 — One-step missing number:

Find the missing number: 8 + □ = 15.

Answer: 15 − 8 = **7**.

Example 2 — Function machine:

A function machine adds 4, then triples the result. If the input is 5, what is the output?

Answer: 5 + 4 = 9; 9 × 3 = **27**.

Example 3 — Two-step missing number:

Find the missing number: 3 × □ + 4 = 19.

Answer: 3 × □ = 19 − 4 = 15, so □ = 15 ÷ 3 = **5**.

Example 4 — Brackets:

Find the missing number: 2 × (□ + 5) = 18.

Answer: □ + 5 = 18 ÷ 2 = 9, so □ = 9 − 5 = **4**.

Example 5 — Multi-step word problem:

I think of a number, double it, then add 7. The answer is 23. What was the number?

Answer: 2 × number + 7 = 23, so 2 × number = 16, so the number = 16 ÷ 2 = **8**.

Foundation

15 questions

1. Find the missing number: 6 + □ = 13.

2. Find the missing number: □ − 5 = 8.

3. Find the missing number: 4 × □ = 24. A) 5 B) 6 C) 7 D) 8

4. Find the missing number: □ ÷ 4 = 5.

5. Find the missing number: 18 − □ = 7.

6. Select ALL the number sentences below where the missing number is 7: A) □ + 3 = 10 B) □ + 6 = 12 C) 2 × □ = 14 D) □ − 2 = 9 E) 21 ÷ □ = 3

7. Find the missing number: □ + 8 = 14.

8. Find the missing number: 5 × □ = 35. A) 5 B) 6 C) 7 D) 8

9. Find the missing number: □ ÷ 4 = 6.

10. Find the missing number: 12 + □ = 19.

11. Find the missing number: □ − 6 = 4.

12. Select ALL the number sentences below where the missing number is 5: A) □ × 2 = 10 B) □ + 7 = 12 C) 15 ÷ □ = 3 D) □ − 3 = 8 E) 6 + □ = 11

13. Find the missing number: 3 × □ = 21.

14. A function machine adds 6, then doubles. If the output is 22, what was the input? A) 5 B) 6 C) 8 D) 11

15. Find the missing number: □ + 12 = 19.

Standard

20 questions

1. A function machine adds 3, then multiplies by 4. If the input is 5, what is the output?

2. A function machine subtracts 2, then doubles. If the input is 8, what is the output?

3. A function machine multiplies by 3, then adds 4. If the input is 5, what is the output? A) 17 B) 18 C) 19 D) 20

4. Find the missing number: 2 × □ + 5 = 19.

5. Find the missing number: □ ÷ 3 + 2 = 9.

6. Select ALL of these where the missing number is 6: A) 4 × □ = 24 B) □ + 9 = 15 C) 20 − □ = 13 D) □ × 5 = 20 E) 2 × □ + 3 = 15

7. Find the missing number: 4 × □ − 3 = 17.

8. A function machine adds 5, then halves. If the input is 7, what is the output?

9. Find the missing number: 3 × □ + 4 = 28. A) 6 B) 7 C) 8 D) 9

10. Find the missing number: □ ÷ 5 − 2 = 3.

11. A function machine triples, then adds 7. If the output is 28, what was the input?

12. Find the missing number: 6 × □ − 4 = 26.

13. Select ALL of these where the missing number is 8: A) □ + 6 = 14 B) 2 × □ = 18 C) 3 × □ − 5 = 19 D) □ − 3 = 5 E) 24 ÷ □ = 3

14. Find the missing number: 5 × □ + 6 = 41.

15. A function machine adds 5, then triples. If the output is 33, what was the input? A) 5 B) 6 C) 7 D) 8

16. Find the missing number: 2 × □ + 7 = 17.

17. A function machine multiplies by 2, then subtracts 4. If the input is 9, what is the output?

18. Find the missing number: □ ÷ 2 + 6 = 11.

19. Find the missing number: 7 × □ − 5 = 23.

20. A function machine adds 9, then halves. If the input is 15, what is the output?

Hard

5 questions

1. Find the missing number: 3 × (□ + 2) = 21.

2. I think of a number, add 5, then triple it. The answer is 24. What was the number?

3. A rectangle has a perimeter of 26cm. If the length is 8cm, what is the width? (perimeter = 2 × (length + width))

4. Find the missing number: 4 × □ + 3 × □ = 21.

5. Two numbers add to 20. One number is 4 more than the other. What are the two numbers?

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